

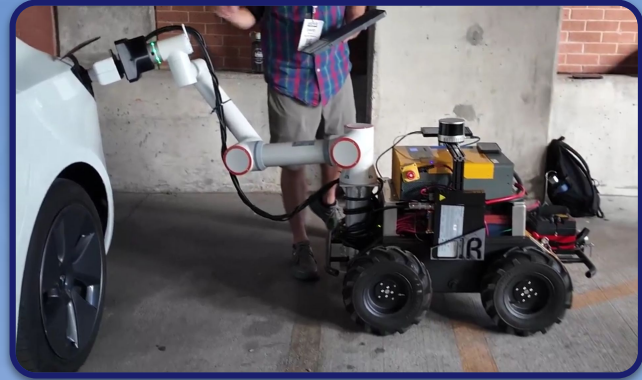
Public Portfolio:
<https://showspace.so/s/jadenfann>

Inductive Robotics project on following slides, not included in public portfolio



Inductive IR3 Prototype

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06/24-09/24



Inductive Robotics Overview

- Current stationary charging infrastructure in parking lots that can only charge 1 car at a time. If a car stays the entire day, the charger will charge the car and then not be used for the rest of the day.
- Inductive proposes an autonomous robotic solution that can charge cars to a certain level, navigate to a charging station, charge another car, etc.
- This would theoretically allow continuous charging flow to all electric vehicles in a parking lot instead of only serving one car per charger for the entire day.



IR3 Project Goals

Design, Prototype, Test first iteration of autonomous EV Charging robot solution

- Requirements Document
- Charging System Components
- Mechanical Design + CAD
- Manufacturing (tolerances, drawings)
- Chassis Modification
- Assembly
- Testing

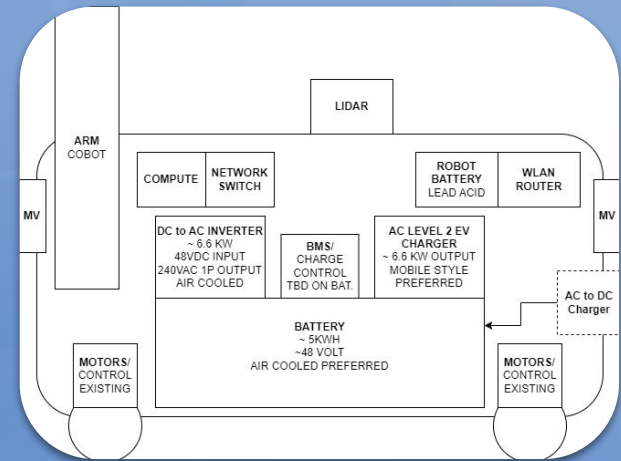


Requirements + Components

- Initial components diagram and requirements document created to guide the development of the project
- Decision matrices created to choose components based on research and requirements

Battery Weighted Decision Matrix:

Battery	Cost	Size	lbs	Cooling	Installation	Energy Capacity	BMS	Chem	Extra Features	
Weights	1	2	2	1	3	1	3	3	1	41
CALB 180AH Battery	1	3	2	2	1	4	2	5		63
EC4# LL-S	3	4	3	4	3	3	5	5		44
LG NMC Proterra Battery Module	5	5	5		1	1	2	3		49
VW ID4 Battery Module	5	5	4	3	1	5	2	3		63
RX-LFP48100 -RU1XU	4	4	3	3	3	3	5	5		45
BatteryEVO Badger	2	3	3		1	4	5	3		49
Commercial Battery (vanguardpower.com)		2	3	3	3	3	5	3		53
APX HV Battery Battery System Growatt	2		2		4	3	5	5	2	



1. Overall Robot

- 1.1 The robot shall fit inside of a single parking space
- 1.2 The robot shall take up the least amount of space behind a parked car as possible
- 1.3 The robot shall be visible to drivers in a parking lot by having a lit feature at least 45 inches above the ground
- 1.4 The robot assemblies modifications to the husky chassis will be designed with maximum reversibility in mind
- 1.5 The robot's center of gravity shall be as low as possible for greater stability
- 1.6 The robot shall be weather resistant to IP22 (eventually IP54)

2. Robot Components

- 2.1 The robot's components shall weigh less than the Husky chassis' maximum load (160lb)
- 2.2 The robot's components shall be securely fixed to the Husky chassis (eventual acceleration/deceleration specifications)
- 2.3 The robot's component assembly shall be modular and easy to manufacture, assemble, maintain, and replace
- 2.4 The robot's components wiring shall meet safety standards for high voltage, etc...

3. Overall Charging System

- 3.1 The charging system shall meet power requirements to charge 50% battery of standard electric vehicles (EVs) in ____ amount of time
- 3.2 **Battery:** 48V, 5KWh power capacity
- 3.3 **DC-AC Inverter:** 8kW, 48V in -> 240Vac, single phase out, (standard plug to charge)
- 3.4 **EV Charger:** 240V interface, 6.6kw or more, (same plug for inverter/adaptor cable)

3.a Battery

- 3.a.2 Battery chemistry shall be lithium iron phosphate or otherwise agreed.
- 3.a.3 Performance:
 - 3.a.3.a Capacity: 4.5 to 6 kWh (made up of 1 or more modules or packs)
 - 3.a.3.b Voltage: 48V nominal
 - 3.a.3.c Output: 6kw continuous @ 25C (made up of 1 more modules or packs)
 - 3.a.3.d Charging amperage: No requirement
- 3.a.3 The battery shall have passive or active air cooling and shall be given sufficient clearance and cooling availability based supplier requirements
- 3.a.4 Compliance: The battery shall be made up of UL 1942 cells tested to one of: UL 2580, UL 1973 or UL 2271
- 3.a.5 Total Battery system weight shall be less than 80 lbs
- 3.a.6 The battery shall be secured with suitable vibration isolation for the expected battery weight

3.b. BMS

- 3.b.1 The BMS shall have the correct specs to control the charging and discharging of the battery with optimal performance
 - For correct battery chemistry
 - Over/under voltage per cell, discharge & charge
 - Over current on charge & discharge
 - Cell balancing
 - Communication via CAN, RS485 or other as agreed

Requirements Summary

Re-designed Clearpath Husky chassis that supports the payload of the IR charging system which includes: Battery for EV charging, Battery Charger for EVs, Inverter or DC charger, Arm, Compute, sensors and more.

2.1 The robot's components shall weigh less than the Husky chassis' maximum load (160lb)

2.2 The robot's components shall be securely fixed to the Husky chassis (eventual acceleration/deceleration specifications)

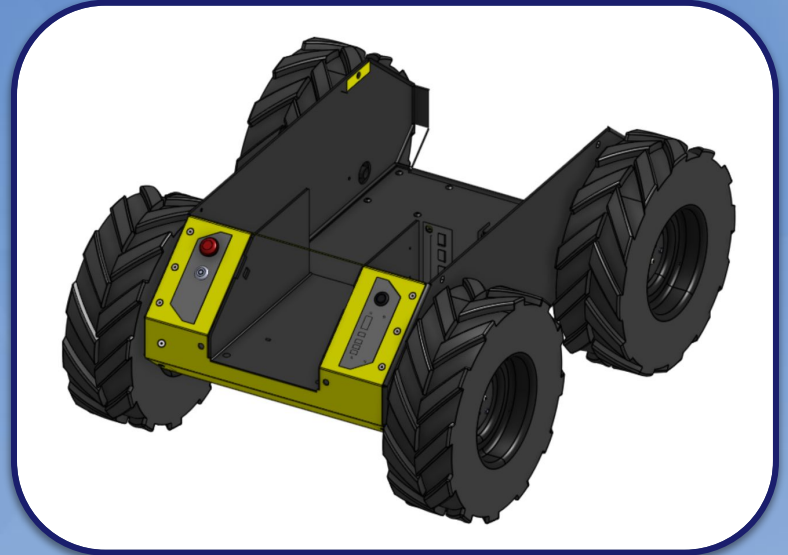
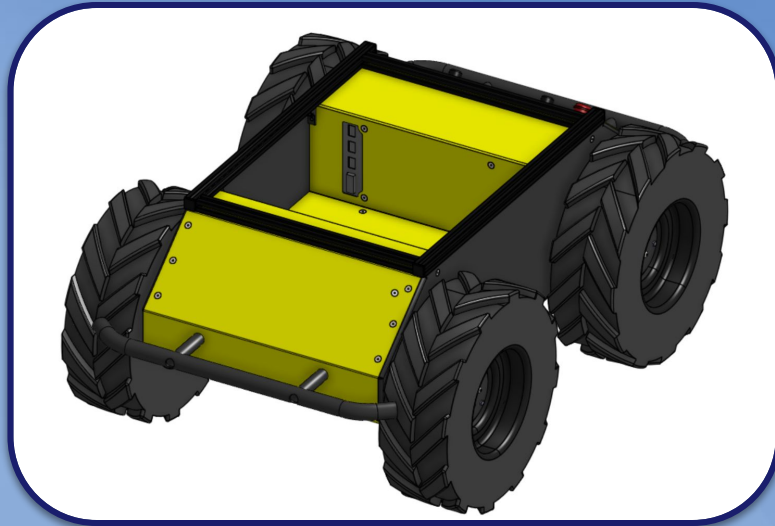
2.3 The robot's component assembly shall be modular and easy to manufacture, assemble, maintain, and replace

2.4 The robot's components wiring shall meet safety standards



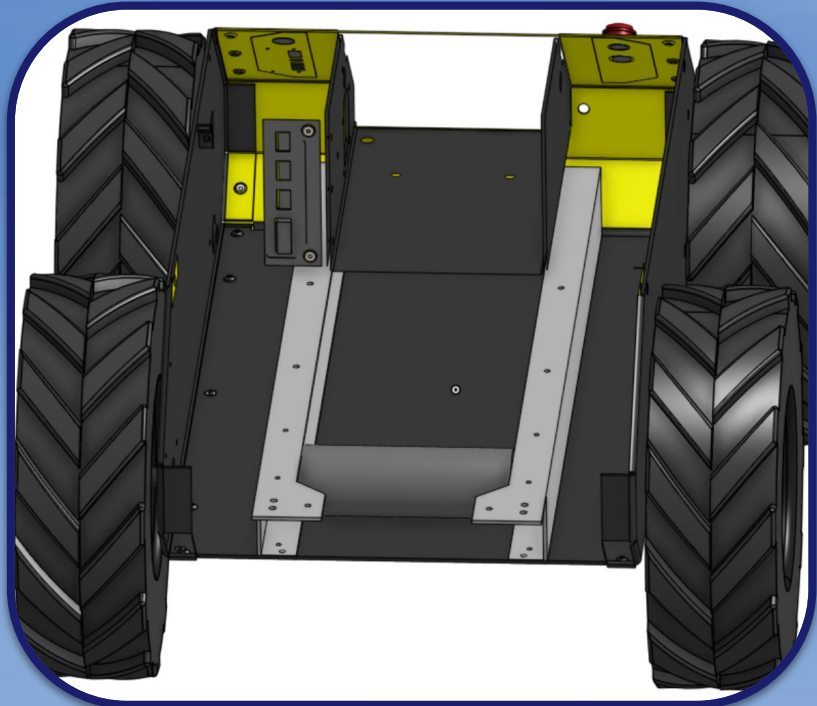
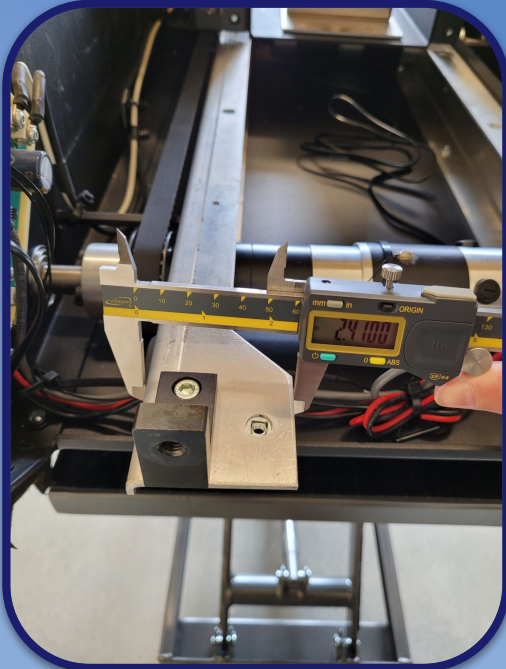
CAD

- Robot design consisted of custom parts and modifications to a Clearpath Husky Unmanned Ground Vehicle used as the drive chassis
- Existing Husky model was modified for further design



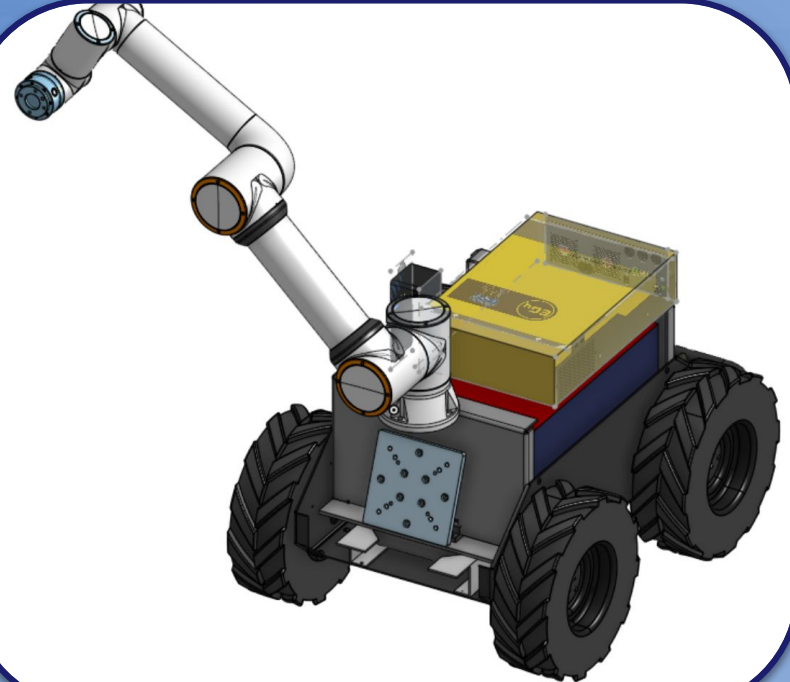
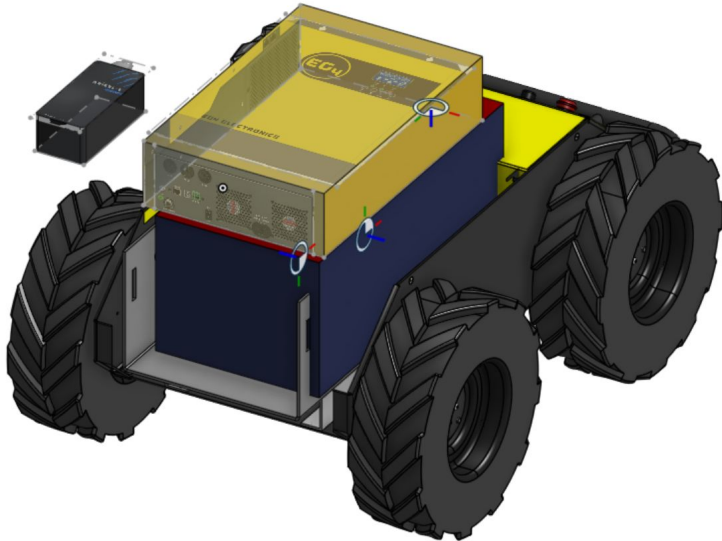
CAD

- Extensive measurement and 3D modeling was required to ensure model accuracy to real world parts

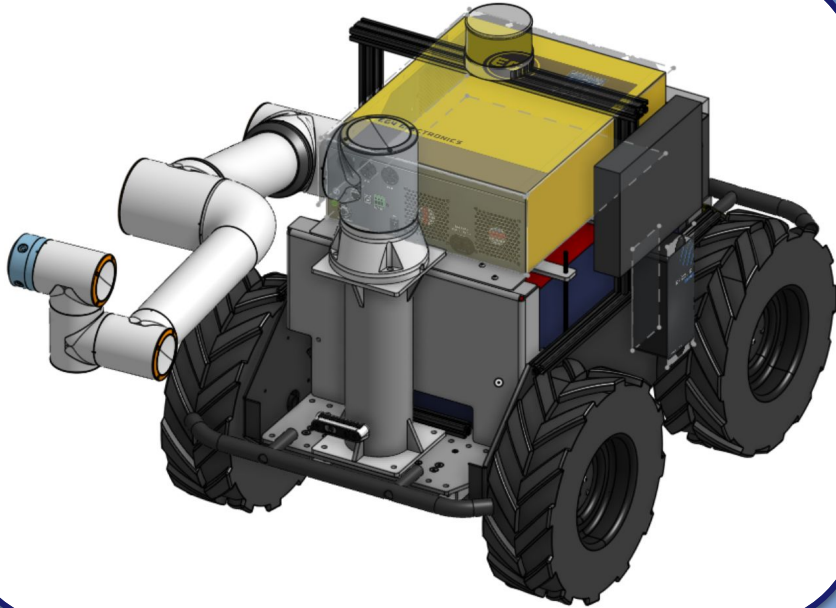


CAD

- Rough CAD Layouts were created from component specifications, keeping in mind mass of parts and center of gravity of entire assembly

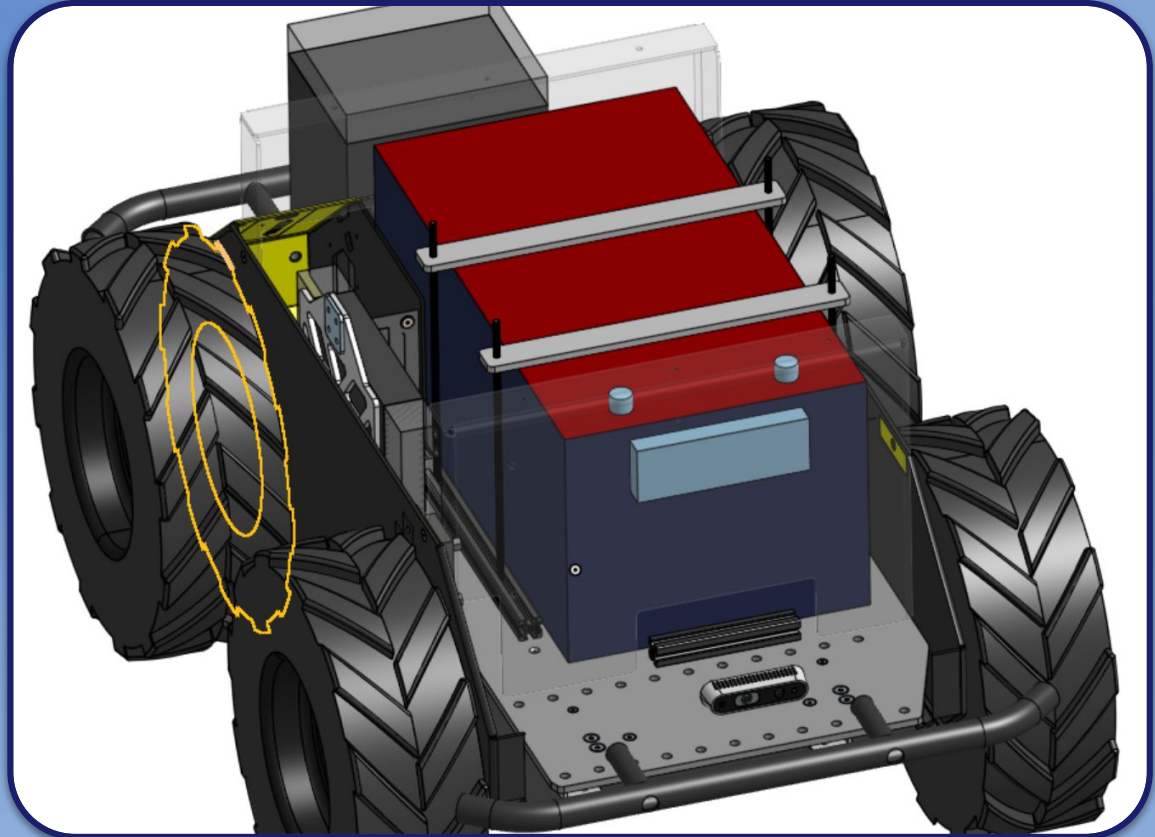


CAD



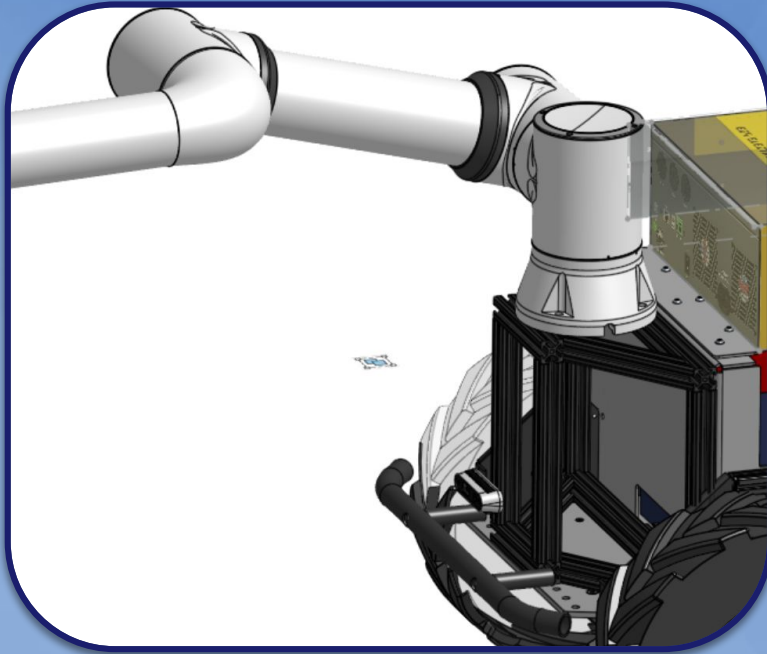
Design Choices – Battery Mounting

Battery secured with aluminum extrusion (x,y) for future flexibility, and threaded rods with steel bars (z). Vibration foam placed around battery.

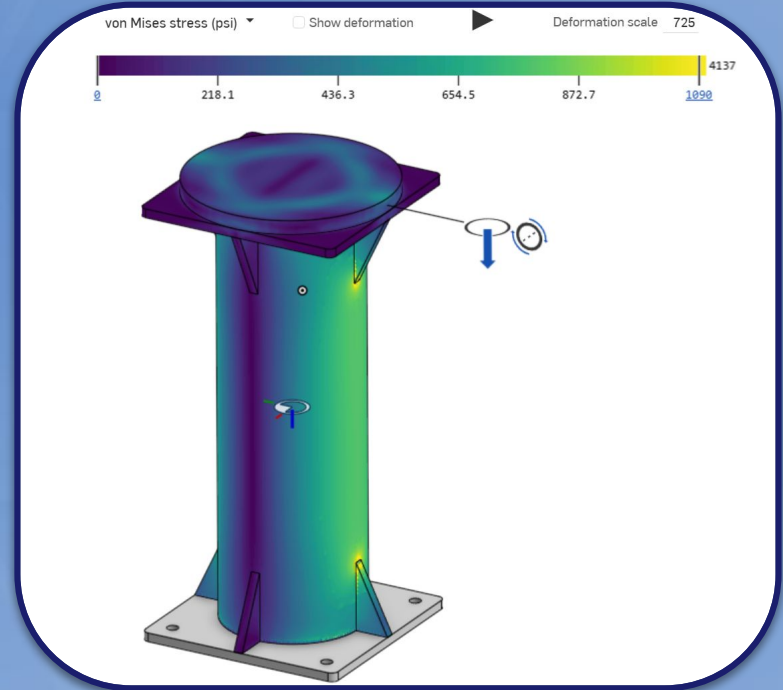
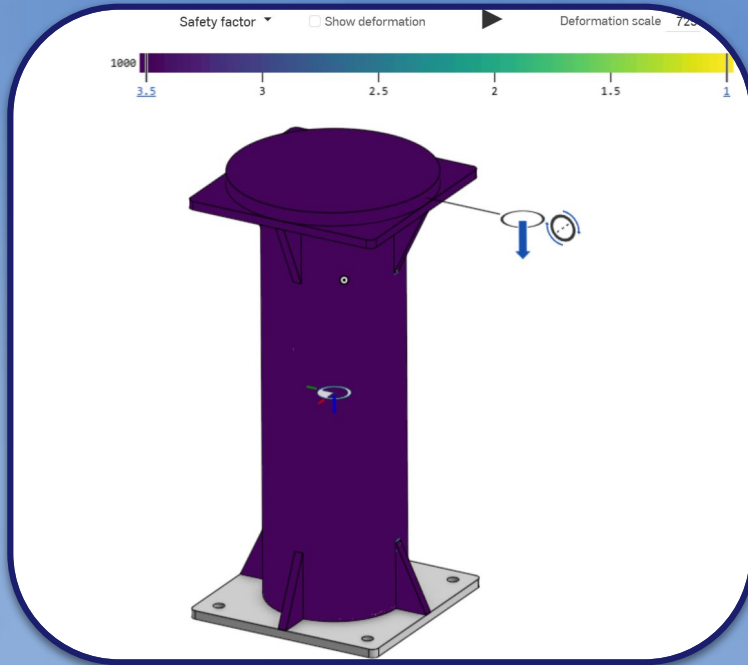


Design Choices – Robot Arm

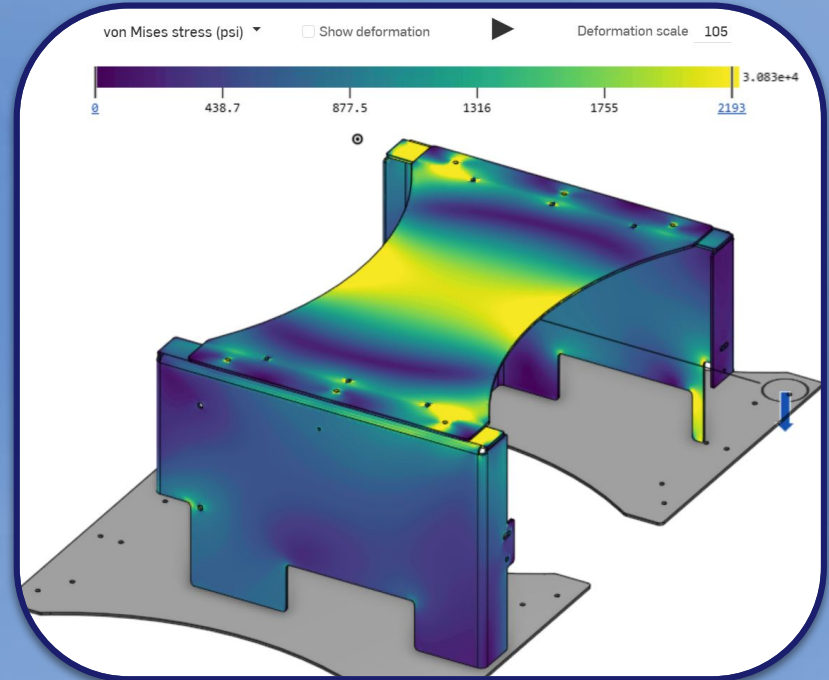
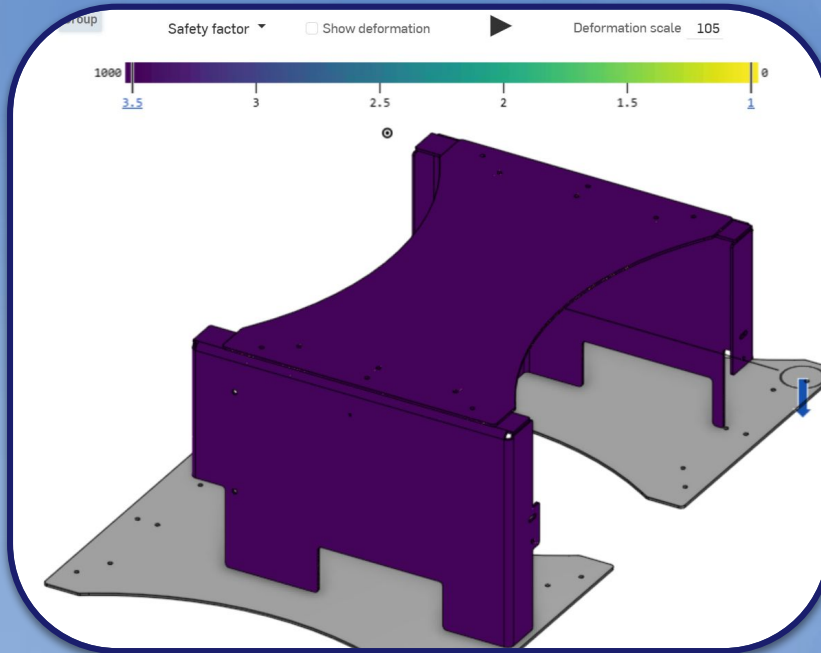
- Welded pedestal selected for simplicity in assembly, reduced complexity, at the expense of future modification flexibility



Validation

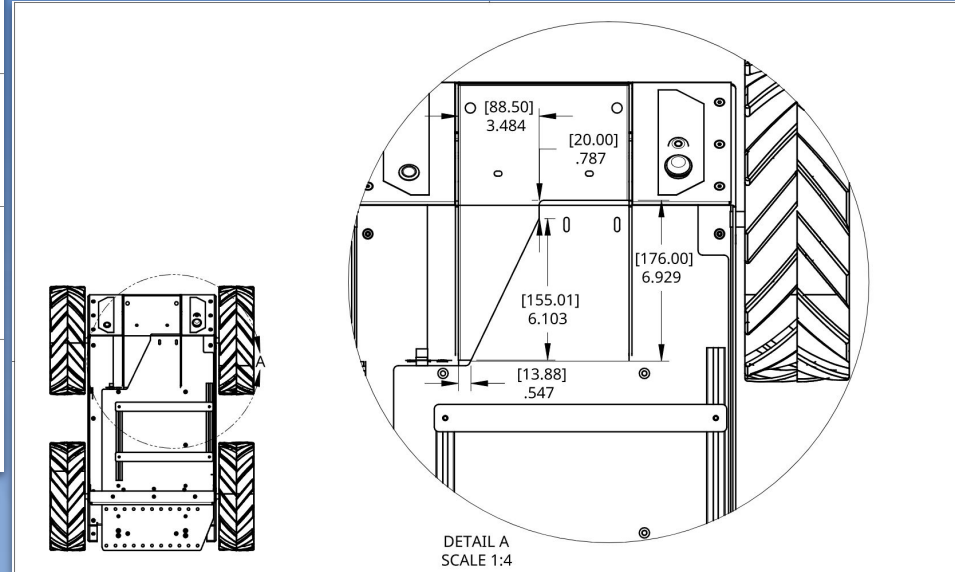
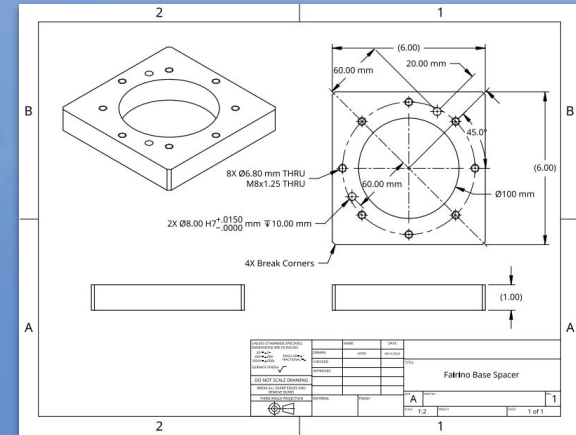
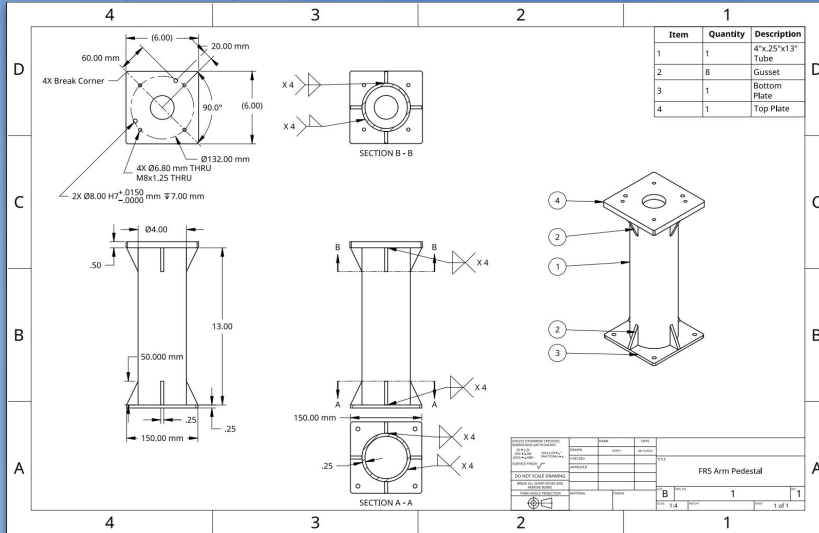


Validation



Manufacturing Drawings

Welded robot arm pedestal, spacer, and approximate cutting diagram for chassis modifications



Challenges

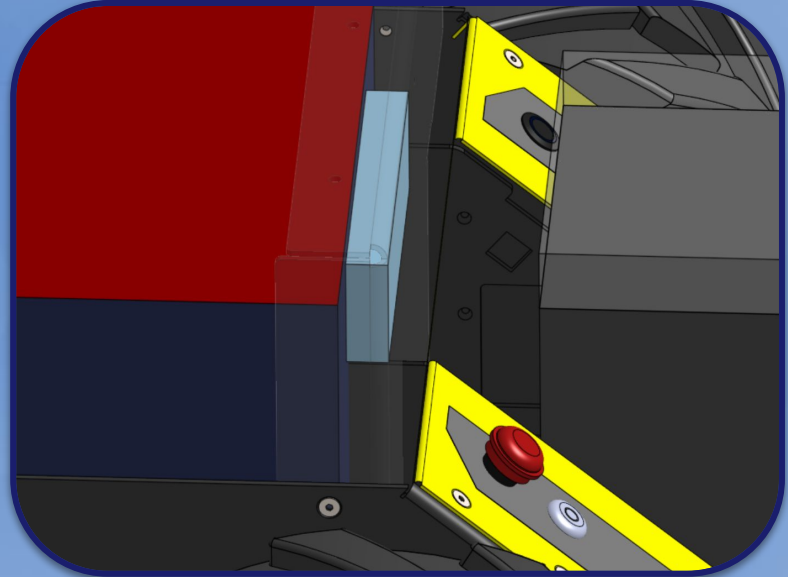
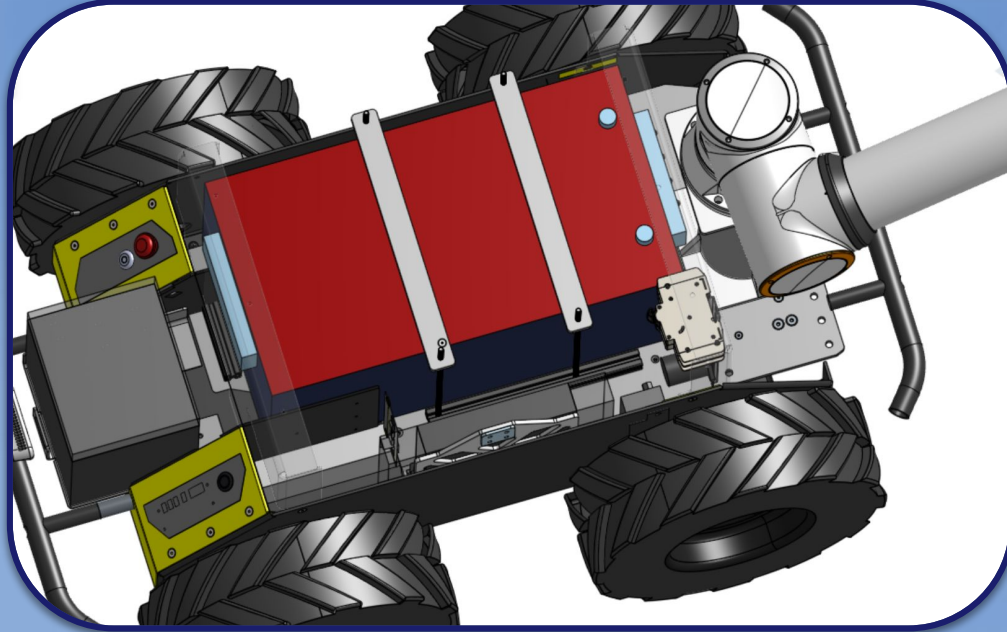
- Model Inaccuracies

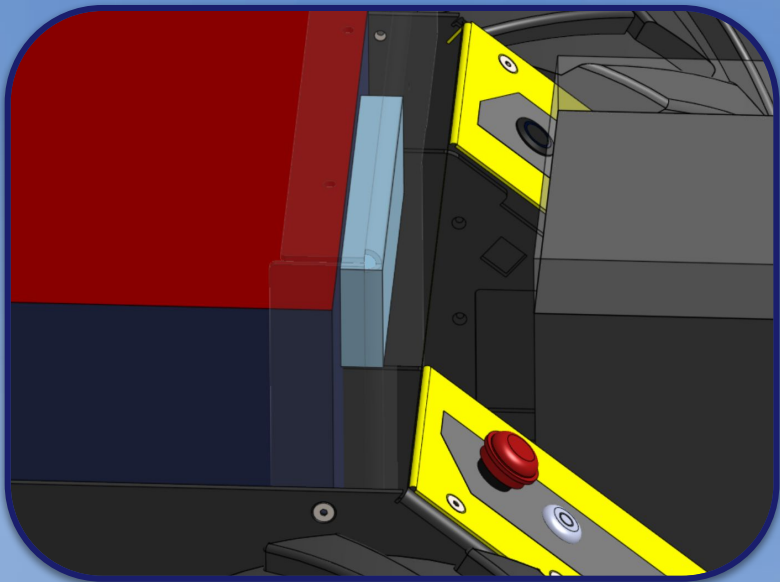
- Many online models and specifications were determined to be inaccurate after acquiring the physical components. Extensive measurements and remodelling was required to ensure ease of manufacturing and assembly

- Assembly

- Designing for assembly, fitting all components into tight spaces and ensuring access for assembly and future maintenance









Final Design Render



Assembly



Assembly



Prototype In Action

